



IOT SMART TUBE MANAGEMENT SYSTEM WITH CLOUD INTEGRATION

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Table of Content

- 1 Introduction
- 2 Problem Statement
- 3 Proposed Solution
- 4 Methodology
- 5 Hardware & Sensors
- 6 Circuit Connections
- 7 Simulation & Results
- 8 Code Overview
- 9 Limitations
- 10 Future Scope
- 11 Conclusion

- **Project Overview:** The IoT Smart Tube Management System is an automated, intelligent faucet controller designed to dynamically regulate water temperature and flow rate.
- **Core Technologies:** Built around the powerful ESP32 microcontroller, utilizing a Proportional-Integral (PI) control algorithm for precise mechanical valve actuation.
- **Cloud Connectivity:** Leverages the Internet of Things (IoT) to transmit real-time telemetry data to the ThingSpeak platform for remote monitoring and historical data analysis.
- **Primary Objective:** To eliminate the physical inefficiencies of manual mixing, prevent accidental scalding through automated hardware safety protocols, and provide users with a centralized digital dashboard.

Problem Statement

- **Inefficient Manual Mixing:** Traditional faucets require manual adjustments to mix hot and cold water, leading to water and energy wastage.
- **Scald Risks:** Sudden spikes in hot water temperature can cause burns, particularly dangerous for children and the elderly.
- **Lack of Real-Time Monitoring:** No mechanisms exist in standard systems to track temperature stability, flow rates, or valve states remotely.
- **No Analytical Data:** Absence of a cloud-based dashboard to visualize system performance and water usage trends over time.

- **ESP32-Based Smart Tube:** An IoT-enabled automated faucet system utilizing an ESP32 microcontroller for real-time control and Wi-Fi connectivity.
- **PI Controller Integration:** Software-based Proportional-Integral (PI) controller to automatically adjust servo valves to match the user's exact temperature setpoint.
- **Automated Scald Protection:** A hard-coded emergency state that instantly cuts off water flow if the mixed water temperature exceeds a critical threshold (50°C).
- **Cloud Integration via ThingSpeak:** Telemetry data (actual temp, setpoints, flow state, servo angles) is pushed to ThingSpeak every 15 seconds for real-time monitoring and analytics.

- ❶ **User Input:** Two potentiometers set the desired Flow Rate and Target Temperature (20°C - 45°C).
- ❷ **Sensor Acquisition:** DHT22 sensor continuously reads the actual mixed water temperature.
- ❸ **Processing Logic:** ESP32 calculates the error (Setpoint - Actual) and runs the PI algorithm to compute the required valve angle.
- ❹ **Actuation:** SG90 Servos physically adjust the 3-way (temperature) and 2-way (flow) valves. An I2C LCD displays local status.
- ❺ **Cloud Transmission:** Data is formatted into an HTTP GET request and transmitted to the ThingSpeak API via Wi-Fi.

Components Used

Processing & Connectivity:

- ESP32 Microcontroller (Wi-Fi Enabled)

Sensors & Inputs:

- **DHT22:** Mixed water temperature monitoring
- **Potentiometer 1 (Temp):** User temperature setpoint
- **Potentiometer 2 (Flow):** User flow rate setpoint

Actuators & Output:

- **Servo 1 (Temp):** 3-way mixing valve control
- **Servo 2 (Flow):** Main flow valve control
- **LCD 16x2 (I2C):** Real-time local status display

Circuit Pin Mapping

Component	ESP32 Pin	Function
DHT22 (Data)	GPIO 21	Actual Temperature Input
Potentiometer 1	GPIO 4 (ADC)	Target Temp Setpoint (20-45°C)
Potentiometer 2	GPIO 5 (ADC)	Flow Rate Setpoint
Servo 1 (Temp)	GPIO 20 (PWM)	Mixing Valve Actuation
Servo 2 (Flow)	GPIO 26 (PWM)	Flow Valve Actuation
LCD I2C (SDA)	GPIO 9	Display Data
LCD I2C (SCL)	GPIO 8	Display Clock

Wokwi Circuit Simulation

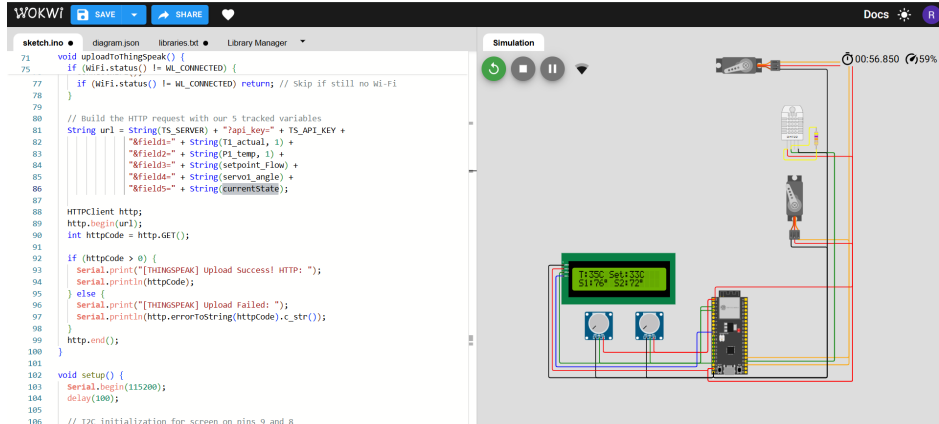


Figure 1: ESP32 Smart Tube PI Controller Circuit executing on Wokwi Simulator

ThingSpeak Cloud Dashboard

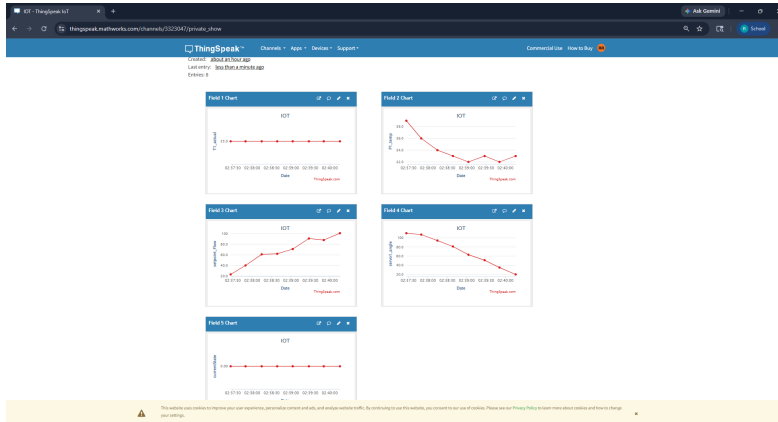


Figure 2: Real-time telemetry showing Actual Temp, Setpoint Temp, Flow, Valve Angles, and System State.

- **PI Control Algorithm:** Calculates positional angle using Proportional ($K_p = 4.0$) and Integral ($K_i = 0.5$) gains with an anti-windup mechanism bound between 0 and 180 degrees.
- **Safety Loop:** If $T1_{actual} \geq 50.0^{\circ}\text{C}$, the system immediately forces `nextState = 2` (EMERGENCY), closing the flow servo entirely and resetting the integral sum.
- **Non-Blocking Cloud Upload:** The control loop evaluates sensor data every 2 seconds, while HTTP GET requests are dispatched to `api.thingspeak.com` asynchronously every 15 seconds to respect rate limits.

- **Sensor Latency:** The DHT22 is a slow-response ambient sensor; in a real-world fluid system, an inline waterproof sensor (like a DS18B20) is required for rapid PI control.
- **Simulated Fluid Dynamics:** The physical delay between valve adjustment and actual temperature change (dead time) is not fully represented by potentiometer adjustments.
- **Internet Dependency:** System telemetry and remote tracking fail if the Wi-Fi connection drops, though local PI control and emergency shutoffs remain active.

- **Hardware Upgrades:** Replace standard servos with robust motorized ball valves suitable for high-pressure municipal water lines.
- **Flow Meters:** Integrate YF-S201 Hall-effect water flow sensors to calculate exact water consumption (liters/minute) for billing and conservation tracking.
- **PID Implementation:** Introduce the Derivative (K_d) term to the control loop to dampen oscillations during rapid target temperature changes.
- **Mobile Application:** Create a companion app using MQTT for instant push notifications during scald alerts instead of relying solely on ThingSpeak polling.

- **Effective Automation:** Successfully engineered a closed-loop system capable of dynamically adjusting water mixture without human intervention.
- **Edge Safety Achieved:** The logic prioritizes local safety over cloud connectivity, guaranteeing immediate flow restriction during dangerous temperature spikes.
- **Cloud Visibility:** Integrated seamlessly with ThingSpeak to provide historical graphing and real-time operational transparency of the complete faucet ecosystem.